置換密碼(佔分 20 分)

問題描述：

瓊斯博士收到亨利博士從南非寄來的一封信，信上的內容如下：

> 親愛的小亨利：
>
> 若要打開失落的法櫃，必須把我給你的一段文字內容將其每個文字依照上帝指引之路轉譯成代碼(由 0-F 組成)；存放在我書房中保險櫃的置換密文表，可以協助你將代碼轉換成密文。最後要打開上帝遺失的法櫃，還需要將密文重新排列後才會是正確的密碼，至於排列規則可以從範例獲得靈感。
>
> 我相信你是那位領受聖靈印記之人。
>
> 亨利一世

上帝指引之路：

I. 將每個文字的第 0 個位元內容進行 NOT 運算後成為代碼的第 0 位元

II. 將每個文字的第 1、3、4、5 位元進行 XOR 運算後成為代碼的第 1 位元

III. 計算每個文字裡的位元內容出現’1’的次數是偶數就填入 0，反之就填入 1 於第 2 位元。

IV. 取出每個文字的第 3 個位元做為代碼的第 3 位元

日記的置換密文表

<table>
<tr><th>代碼</th><th>密文</th><th>代碼</th><th>密文</th><th>代碼</th><th>密文</th><th>代碼</th><th>密文</th></tr>
<tr><td>0</td><td>$#</td><td>4</td><td>$@[</td><td>8</td><td>#<<</td><td>C</td><td>$[</td></tr>
<tr><td>1</td><td>|[</td><td>5</td><td>!<</td><td>9</td><td><|></td><td>D</td><td>@<#</td></tr>
<tr><td>2</td><td>>#[</td><td>6</td><td>#|</td><td>A</td><td>][</td><td>E</td><td>$>!</td></tr>
<tr><td>3</td><td>>]]</td><td>7</td><td>^@!</td><td>B</td><td>^^$</td><td>F</td><td>@$</td></tr>
<tr><td colspan="8">密文符號優先權，由最左的最高優先權排至最右的最低優先權：| ^] [@ > < $ # !</td></tr>
</table>

輸入說明：

一段文字內容由英文字母、數字和符號組成的字串 L，其字串長度為 1≤L≤500。

輸出說明：

對於每根據轉譯後密文內符號出現次數決定該密文符號的輸出順序，若是遇到相同出現次數則依照密文符號優先順序排之。

範例 1：

輸入：

Proper preparation solves 80% of life’s problems.

輸出：

]|[#><!^@$

範例 2：

輸入：

Every noble work is at first impossible.

輸出：

<#[!>|^]@$

Permutation Cipher (20 Points)

Problem Description:

Dr. Jones received a letter from Dr. Henry sent from South Africa. The content of the letter is as follows:

> Dear Junior Henry,
>
> To open the Ark of the Covenant, you must translate each character of a text I gave you into a code (composed of 0-F) according to the Road of God's Guidance. The Permutation Cipher Table kept in the safe of my study can assist you in converting the codes into ciphertexts. Finally, to open God's lost Ark, you need to rearrange the ciphertexts to form the correct password. As for the rearrangement rules, you can get inspiration from the examples.
>
> I believe you are the one who received the seal of the Holy Spirit.
>
> Henry I

The Road of God's Guidance:

I. Perform a NOT operation on the 0th bit of each character to become the 0th bit of the code.

II. Perform an XOR operation on the 1st, 3rd, 4th, and 5th bits of each character to become the 1st bit of the code.

III. Count the number of times the bit content '1' appears in each character. If it is an even number, fill in 0; otherwise, fill in 1 for the 2nd bit.

IV. Take the 3rd bit of each character as the 3rd bit of the code.

The Diary's Permutation Cipher Table:

<table>
<tr><th>Code</th><th>Ciphertext</th><th>Code</th><th>Ciphertext</th><th>Code</th><th>Ciphertext</th><th>Code</th><th>Ciphertext</th></tr>
<tr><td>0</td><td>$#</td><td>4</td><td>$@[</td><td>8</td><td>#<<</td><td>C</td><td>$[</td></tr>
<tr><td>1</td><td>|[</td><td>5</td><td>!<</td><td>9</td><td><|></td><td>D</td><td>@<#</td></tr>
<tr><td>2</td><td>>#[</td><td>6</td><td>#|</td><td>A</td><td>][</td><td>E</td><td>$>!</td></tr>
<tr><td>3</td><td>>]]</td><td>7</td><td>^@!</td><td>B</td><td>^^$</td><td>F</td><td>@$</td></tr>
<tr><td colspan="8">Ciphertext symbol priority, ordered from the highest priority on the far left to the lowest priority on the far right: | ^] [@ > < $ # !</td></tr>
</table>

Input Description:

A text string L consisting of English letters, numbers, and symbols, where the string length is $1 \le L \le 500$.

Output Description:

For each case, determine the output order of the ciphertext symbols based on the frequency of occurrence of the symbols within the translated ciphertext. If encountering the same frequency of occurrence, arrange them according to the ciphertext symbol priority order.

Example 1:

Input:

Proper preparation solves 80% of life’s problems.

Output:

]|[#><!^@$

Example 2:

Input:

Every noble work is
at first impossible.

Output:

<#[!>|^]@$